Qiyang Hong, Ph.D.

Biomedical AI Researcher

Machine Learning for Disease Risk & Progression

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Google Scholar: https://scholar.google.com.hk/citations?user=1PCtyx8AAAAJ&hl=zh-CN

EDUCATION

- **Ph.D. in Biomedical Engineering**, Tsinghua University & Peking Union Medical College, Beijing, China (Sept 2023–June 2026)

  Advisor: Prof. Erping Long

  Degree awarded: Doctoral Degree in Engineering (June 26, 2026)

  Thesis: Construction of the ukbFound Foundation Model for Deep Phenotyping Data and Its Applications in Disease Stratification, Multimorbidity Networks, and Risk Prediction

  Thesis defense: May 25, 2026
- **M.S. in Biochemistry and Molecular Biology**, Xiamen University, Xiamen, China (Sept 2014–June 2017)

  Advisor: Prof. Ning-Shao Xia

  Research contribution: Bioinformatics support, data processing, and computational analysis.
- **B.Eng. in Bioengineering**, Jimei University, Xiamen, China (Sept 2010–July 2014)
- **B.Eng. in Computer Science and Technology,** The Open University of China, Beijing, China (Sept 2023-Jan 2026)

SELECTED PUBLICATIONS

*Equal contributor; #Corresponding author.

2026

1. **Hong Q**[*], Wang C[*], ..., Pang J#, Long E[#]. A foundational model encodes deep phenotyping data and enables diverse downstream applications. *npj Digital Medicine*. 2026;9:576. doi:10.1038/s41746-026-02736-w.

2. Sun X[*], **Hong Q**[*], ..., Wan P[#], Long E[#]. Model confrontation and collaboration: A debate intelligence framework for enhancing medical reasoning in large language models. *Cell Reports Medicine*. 2026;7:102547. doi:10.1016/j.xcrm.2025.102547.

3. Yi E[*], Cui J[*], Wang H[*], Wu F*, **Hong Q**[*], ..., Long E[#], Ran P[#]. Multiomics Mendelian randomization identifies serpin family G member 1 as a chronic obstructive pulmonary disease modulator. *Signal Transduction and Targeted Therapy*. 2026;11:34. doi:10.1038/s41392-025-02547-7.

4. Pang J[*], Ren X[*], **Hong Q**[*], Huang K, Wang J, Li D, Huang X, ..., Liang C[#], Long E[#], Wang J[#], Yang T[#], Wang C. Proteo-metabolomic insights into the progression of chronic obstructive pulmonary disease and lung function decline. *Respiratory Research*. 2026 (online ahead of print). doi:10.1186/s12931-026-03768-2.

5. Wu W[*], Zhou M[*], Chen J, Wang J, Hong Q, ..., Pang J[#], Wang J[#], Long E[#]. Deep multi-omics profiling reveals three molecular subtypes of chronic obstructive pulmonary disease in a unique biomass-exposed Chinese population. *Med*. 2026;7:101033. doi:10.1016/j.medj.2026.101033.

2025

6. Long Y[*], Ni X[*#], Chen T[*], Huang X, Hong Q, ..., Zhang T[#], Long E[#]. Genetic determinants of gene expression noise and its role in complex trait variation. *Cell Reports*. 2025;44:116612. doi:10.1016/j.celrep.2025.116612.

2024

7. Lin K[*], **Hong Q**[*], Fu Y*, ..., Chen M[#]. Cervical HPV infection and related diseases among 149,559 women in Fujian: an epidemiological study from 2018 to 2023. *Frontiers in Microbiology*. 2024;15:1418218. doi:10.3389/fmicb.2024.1418218.

TECHNICAL SKILLS

- **Biomedical Data Science & Machine Learning:** Machine learning, multimodal and longitudinal data modeling, model interpretation, and analysis of large-scale biomedical and clinical datasets.
- **Bioinformatics & Statistical Analysis:** Multi-omics, genomic and transcriptomic data analysis; survival analysis, multiple-testing correction, and resampling methods.
- **Programming & Data Analysis:** Python, PyTorch, NumPy, pandas, scikit-learn, and reproducible data-analysis workflows.

PROFESSIONAL EXPERIENCE

- **Group Leader, Molecular Diagnostics**
  Affiliated Hospital of Putian University, Prenatal Diagnosis Center
  Aug 2020 – Jun 2023
- **Bioinformatics Supervisor**
  Berry Genomics Co., Ltd.
  Jul 2017 – Aug 2020

TEACHING EXPERIENCE

- **Part-time Course Tutor**
  Experimental College, The Open University of China
  Sept 2023 – Aug 2025
  Course: Special Topics in Artificial Intelligence

PEER REVIEW EXPERIENCE

Invited reviewer (2026; 5 invitations): Journal of Medical Internet Research (2); JMIR AI (1); JMIR Medical Education (1); JMIR Cardio (1).

Co-reviewer (with Prof. Erping Long): Nature Medicine; Nature Biomedical Engineering; Frontiers in Aging Neuroscience.

SELECTED RESEARCH PROJECTS

- **ukbFound for patient stratification and disease risk prediction**
  Developed and evaluated a machine-learning model using deep phenotyping data from more than 500,000 UK Biobank participants. The study explored disease risk prediction and multimorbidity, and identified distinct patient subgroups in 289 diseases (npj Digital Medicine, 2026).
- **Evaluation of medical language models**
  Contributed to data acquisition, curation, and analysis for evaluating collaborative language models on medical questions and clinical reasoning tasks (Cell Reports Medicine, 2026).
- **Multi-omics analysis of COPD and lung function decline**
  Contributed to studies integrating genetic, protein, metabolite, and clinical data to investigate chronic obstructive pulmonary disease (COPD), molecular subtypes, and longitudinal lung function decline (Signal Transduction and Targeted Therapy, Med, and Respiratory Research, 2026).